MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING

MODULE 2 UNIT 1
Lesson Video 3 Transcript

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Multi-agent systems

JACOB ANDREAS: Let's conclude by talking about a couple of current research topics in the design of agentic systems.

To begin with, there's an enormous amount of interest right now in multi-agent systems. Often, we can think of these multi-agent systems as including some sort of controller agent that is delegating subtasks to individual subagents, which might now be tools that are not implemented as pieces of code, but that are instead themselves entire language models, or language models with their own external scaffolding, that can accomplish particular subtasks.

In most cases, it's often useful to think of these multi-agent systems as just single agents in which parts of the computation or parts of the planning work are being done by independent subsystems rather than in a single thread.

That being said, one use case where it really does make sense to think about multi-agent systems and interactions between multiple agents are large distributed organizations where individual users might have access to their own agent that has access to those users' personal information, and where the agents are being put in charge of deciding what information to share or not to share, such that behavior can be coordinated along many groups of agents.

The usual information security precautions that we talked about earlier in this lesson still apply. We need to make sure that users sign off on any data that their agent shares on their behalf. But with this kind of architecture, user-specific agents can do most of the work of figuring out what to ask users, what kind of information to share, and how to coordinate with other agents – without necessarily requiring a lot of hands-on supervision or approval.

Multimodality

ANDREAS: So far, when we've been looking at interactions with agents, those interactions have taken place entirely in text. Users have sent text to agents as input, and agents have produced text as output – either taking the form of programs to be run, or textual messages to send to users. But we can also build agents – thinking back to our original "taking out the trash" example – that take actions in the real world of sight and sound and motion.

Nowadays, most language models that we build agents on top of are actually multimodal. You can give them images as part of their input. They can parse these images the same way they might parse text; they can sometimes even generate images as output.

And so, if we want to support things like customers taking pictures of objects and asking agents questions about them, this is easy. And of course, once you have a model that can look at images and can write code, that takes actions that cause side effects in the real world, now we can put a language model agent on a robot and have the language model actually driving that robot around in the real world. And again, this is an idea that goes back really to the earliest days of the field, long before the current generation of LLMs. But with LLM-powered agents, you can build some really surprisingly effective robotic demos.

It's worth noting that the best of these systems are certainly not pure-language models. Typically, to build a language-model-powered robot, we prompt the language model to write really high-level programs that then get executed by much lower-level robot controllers that are implemented as smaller neural networks that have been specially trained to perform single operations like grasping or pulling or moving.

As a result, we're still very far away from being able to perform robotics tasks with language models that require both really complicated reasoning or planning, and really fine-grained manipulation or perception of the world.

Conclusion

ANDREAS: So, to wrap up, a language-model agent is just a language model equipped with the ability to perceive and interact with the external virtual or physical world.

Building an agent requires us to think about observation, action, planning, memory, and safety. In the coming years, we should expect lots of changes in the kinds of capabilities that language-model agents exhibit. First of all, we should expect that their ability to write complex programs and solve complex planning problems will continue to increase rapidly, while maybe we'll see slower progress on both some of these safety issues and real-world vision and control.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.